

Ming Huang

Mechanical Engineer | Robotics R&D

mh2668@cornell.edu | [mh2668-droid.github.io/Portfolio](https://github.com/mh2668-droid) | linkedin.com/in/minghuang-mecheng

EDUCATION

Cornell University, Ithaca, NY

Aug 2025 - Present

- M.S. Mechanical Engineering (Robotics), concentration in Dynamics and control

Michigan State University, East Lansing, MI

Sep 2021- May 2025

- B.S. Mechanical Engineering, concentration in Biomedical Engineering
- GPA: 3.72/4.00

RESEARCH & ENGINEERING EXPERIENCE

Collective Embodied Intelligence Laboratory, Department of Electrical and computer Engineering, Cornell University

Robotics R&D – Advisor: Dr. Kirstin H. Petersen

Aug 2025 – Present

- Leading and building a fully autonomous aquatic robot for invasive vegetation management.
- Currently integrating an underwater manipulator end-effector for autonomous detection, cutting, and collection of submerged vegetation.

Prism Controls (PMSI)

Mechanical Engineering Intern

May 2024 - Aug 2024

- Independently developed BarnFlowDynamics, a large-scale CFD framework for airflow and heat transfer analysis in agricultural facilities using OpenFOAM.
- Built and executed >100M cell parallel simulations, configuring turbulence and thermal transport models on HPC systems.
- Led end-to-end simulation workflow design, including meshing, solver configuration, parallelization, and convergence optimization.

Neural Engineering Laboratory, Department of Mechanical Engineering , Michigan State University

Research Assistant – Advisor: Dr. Galit Pelled

Aug 2024 – Oct 2025

- Conduct research on neural circuitry and motor control in octopus arms to understand decentralized learning and adaptation.
- Designed and executed stimulation experiments using high-density multi-electrode arrays.
- Developed analysis pipelines for signal processing, spike detection, and neural data visualization.

Computational Biomechanics Laboratory, Department of Mechanical Engineering, Michigan State University

Research Assistant – Advisor: Dr. Lik Chuan Lee

Aug 2024 – May 2025

- Developed finite element models of swine hearts to study ventricular–aortic interactions and cardiac mechanics
- Built and refined 3D meshes from echo data, defined myocardial fiber architecture, and prepared datasets for electromechanics simulations.
- Performed waveform analysis and validation, aligning simulated pressure volume loops with experimental swine data.

PUBLICATIONS AND MANUSCRIPTS

1. Ziaei-Rad, V., **Huang, M.**, Lee, L.-C. *Mechanisms of Reduced Global Longitudinal Strain in Heart Failure with Preserved Ejection Fraction: Insights from a Novel Swine Model and Computational Framework*. manuscript in preparation.
2. **Huang, M.**, Pelled, G. *Investigating Neural Plasticity in Octopus Arm Tissue via Electrical Stimulation*. Poster presentation.
3. **Huang, M.**, Wang, J., Yu, L., Feng, R., He, Y. (2021). *Using speech recognition to analyze the physical condition of the elderly and its application in telemedicine*. *International Core Journal of Engineering*, 7(12), 619–626.
DOI:10.6919/ICJE.202112_7(12).0086

Teaching Experience

Teaching Assistant – MAE2250: Intro to Mechanical Design, Cornell University

Jan 2026 – present

- Operate lab sections and helping students with CAD, prototyping, and mechanical design project.
- Guide students through the design process, troubleshooting fabrication issues and manufacturing planning.

HONORS AND AWARDS

- Dean's List at MSU *2021-2025*
- 5th place in World First Robotics Competition in Midwest Region *2017-2018*

TECHNICAL SKILLS

- **Robotics:**
Robot Control, Embedded sys, actuator control, sensor integration
- **Hardware & Embedded Sys.:**
Microcontroller, LabVIEW, Pspice
- **Software:**
Python, C++, MATLAB, ROS, Gazebo, Linux, Git
- **Design and Fabrication:**
CAD(Fusion 360, SolidWorks, NX), Machining, 3d printing
- **Computational Modeling:**
CFD, FEA